F78PB Probability and Statistics B Practical Test 2022

You have one hour to complete this test. There are two questions and the results contribute 10% towards your final grade for the course. Type the answers into a Word document and copy and paste any graphs and diagrams that help illustrate your work. At the end, submit the file through Canvas. Go to “Assignments” and select your campus.

You can obtain the datasets by downloading the file test.RData in Canvas. When you open R, go to File → Load Workspace and load this file. This will give you the data you need to work with. Alternatively, you can use the scan command to copy and paste the data.

1. The results below arise from taking 50 samples of size 20 from a population with known variance 100.

For each sample, the variance has been calculated using two different estimators.

estimator1 = $\sum_{i=1}^{i=n} \frac{(x_i-\bar{x})^2}{n}$ and estimator2 = $\sum_{i=1}^{i=n} \frac{(x_i-\bar{x})^2}{n-1}$

Variances calculated by estimator 1

70.1 83.0 124.7 93.0 73.0 218.3 99.1 128.6 87.4 80.7 67.7 59.5

158.1 79.2 127.3 94.4 74.4 73.3 85.5 51.0 69.4 104.7 84.0 215.9

78.6 98.1 56.7 76.3 97.8 103.0

Variances calculated by estimator 2

72.5 85.9 129.0 96.2 75.5 225.8 102.5 133.0 90.4 83.5 70.0 61.5

163.6 81.9 131.7 97.7 77.0 75.8 88.4 52.8 71.8 108.3 86.9 223.3

81.3 101.5 58.7 78.9 101.2 106.6

These data are in the R vectors called *estimator1* and *estimator2*.

By carrying out appropriate calculations on R, state which of the two estimators gives a better estimate of the population variance. Explain briefly why your answer supports the usual recommended way of calculating sample variances.

[Total 4 marks]

2 Suppose that X = (X_1,X_2,...,X_{50}) is a random sample from an Exp(1/θ) distribution

with mean $\theta > 0$. Let $X_{(1)} \equiv \min\{X_1,X_2,...,X_{50}\}$. We will consider the sampling distributions of two possible estimators of the value of θ, given by

$$\hat{\theta}_1 = \bar{X} \quad \text{and} \qquad \bar{\theta}_2 = 50X_{(1)}$$

(a) Simulate 10,000 samples from an Exp(1/4) distribution. For each sample, calculate the value for each estimator of θ. You might like to use the following code:

```
theta.1=rep(0,10000)
theta.2=rep(0,10000)
for (i in 1:10000){
x=rexp(50,0.25)
theta.1[i]=mean(x)
theta.2[i]=50*min(x)
}
```

Present appropriate summary statistics for the distributions of theta.1 and theta.2.

(3 marks)

(b) Plot histograms of the sampling distributions of the two estimators and very briefly describe their shapes. (3 marks)

(c) For your particular simulation, use R to estimate the bias and mean squared error (MSE) of each estimator.

(3 marks)

(d) The following sample results are believed to come from an Exp(θ) distribution. Estimate the value of θ, explaining your choice of method. The data are in the R vector *testresults*.

15.95 11.85 7.65 3.05 1.65 3.05 2.05 4.95 5.35 5.95 0.15 1.45
0.25 0.45 1.55 13.25 9.45 1.55 4.35 2.35 11.35 3.85 23.05 8.85
3.85 14.05 36.55 13.15 5.75 10.75 0.65 0.75 1.95 9.55 1.25 32.65
0.65 10.45 6.95 19.95 0.01 2.45 7.45 18.85 19.45 11.95 12.05 11.45
5.45 0.01

(3 marks)

[Total 12 marks]